Benchmark Method Specifications

Framework	Core Parameters	Specification / Baseline Setup
HERA	Threshold Bootstrap (B = 2000)	Sequential Hierarchical (M1 -> M2 -> M3); Holm-Bonferroni FWER control (alpha = 0.05)
TOPSIS	Criteria Weights: w = [0.40, 0.40, 0.20]	Vector Normalization; Euclidean distance to ideal solutions; relative closeness ranking
Borda Count	Positional Rank Sum (Equal weights across M1-M3)	Sum of individual metric ranks; consensus ranking by lowest rank sum
Wilcoxon-Copeland	Pairwise Tournament (Net wins minus losses)	Paired Wilcoxon test with step-down Holm-Bonferroni FWER control (alpha = 0.05)

Experimental Benchmark Scenarios (24 Conditions)

Benchmark Suite	N Conditions	Candidates (N)	Sample Size (n)	Noise (σ, %)	Effect Size (Cliff's d [Range])	Correlation (ρ)
Core	15	10, 12, 14	50	2%, 4%, 6%, 8%, 10%	d ≈ 0.50 [0.45 - 0.60]	0
Sample Size Sensitivity	6	10, 12, 14	25, 100	4%	d ≈ 0.50 [0.45 - 0.60]	0
Effect Magnitude Sensitivity	2	12	50	4%	d ≈ 0.25 [0.20 - 0.30], d ≈ 0.80 [0.75 - 0.90]	0
Correlation Sensitivity	1	12	50	4%	d ≈ 0.50 [0.45 - 0.60]	0.5

Ground-Truth Benchmark Calibration

Candidate Hierarchy (N = 10)

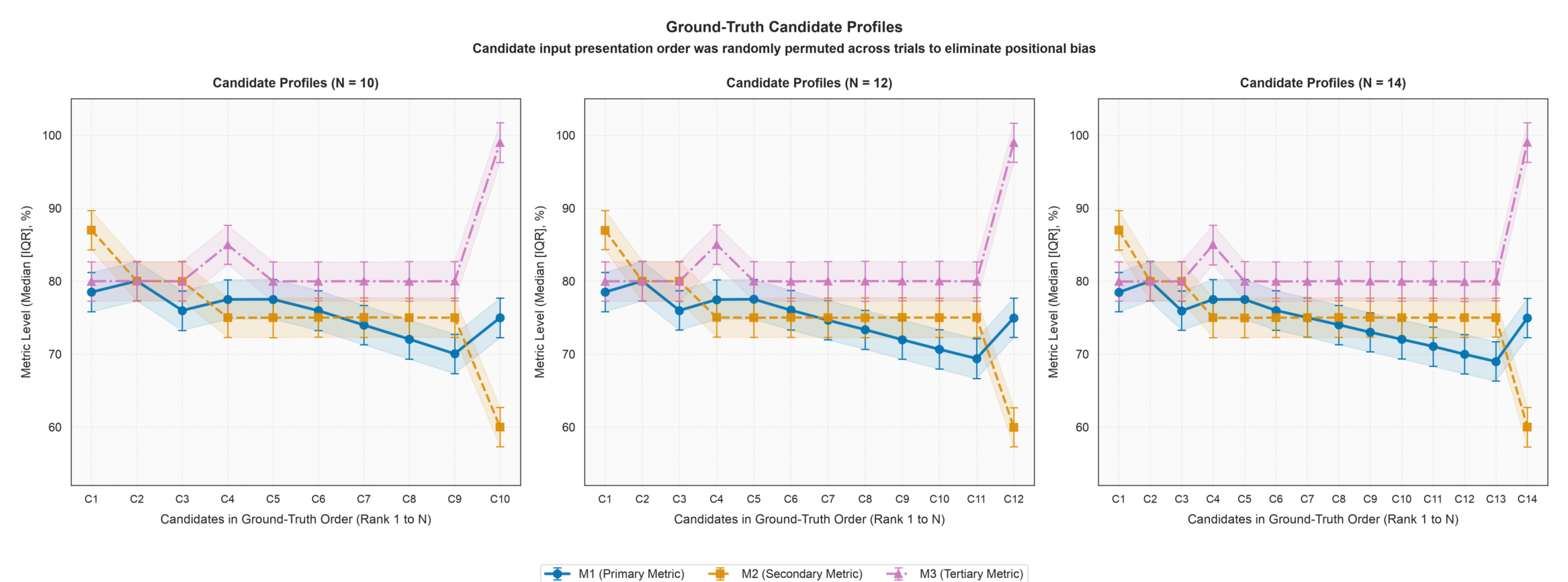
Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	74.0%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	70.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

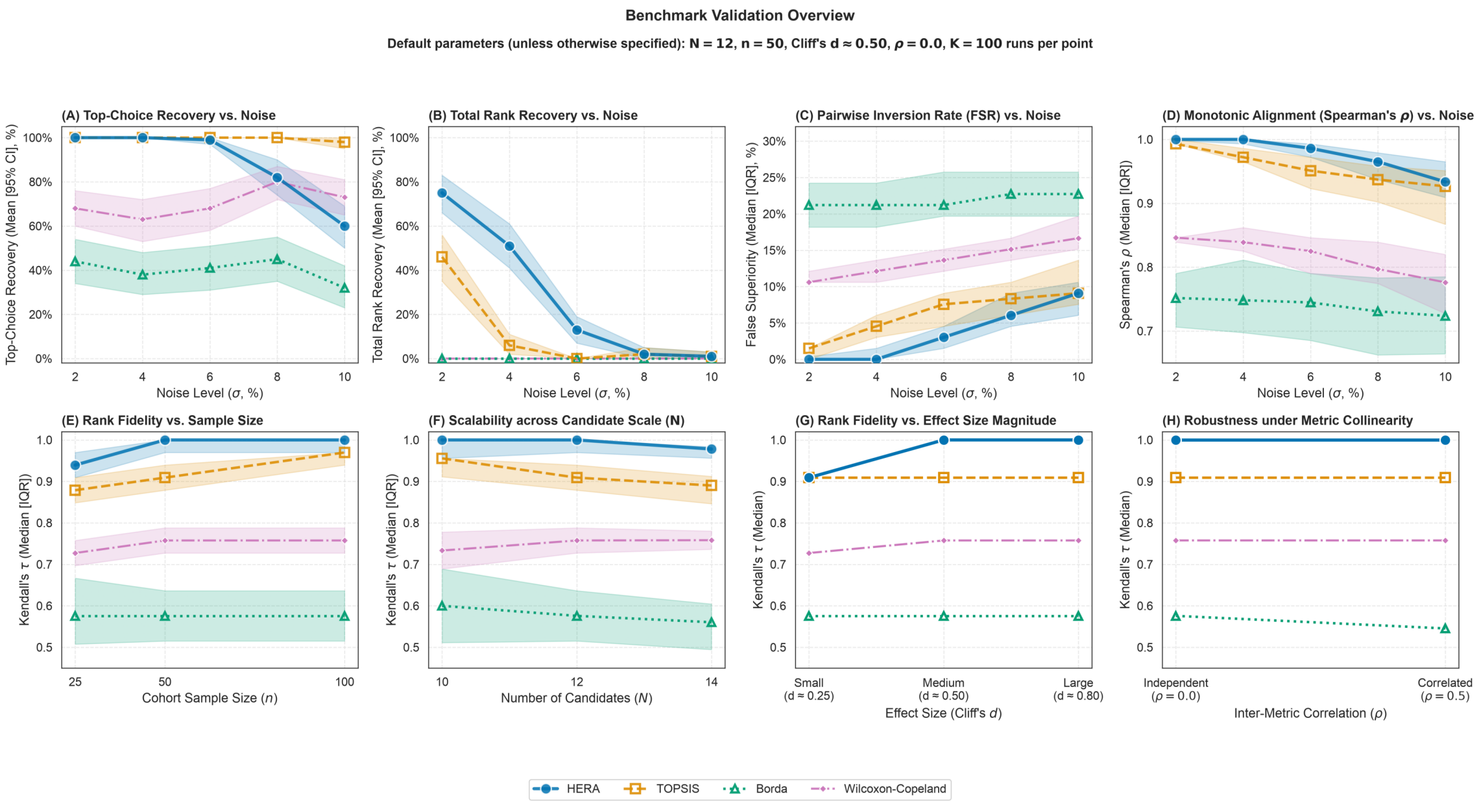
Candidate Hierarchy (N = 12)

Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	74.7%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	73.3%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	70.7%	75.0%	80.0%	Monotonic Baseline Step
C11 (Rank 11)	69.3%	75.0%	80.0%	Monotonic Baseline Step
C12 (Rank 12)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

Candidate Hierarchy (N = 14)

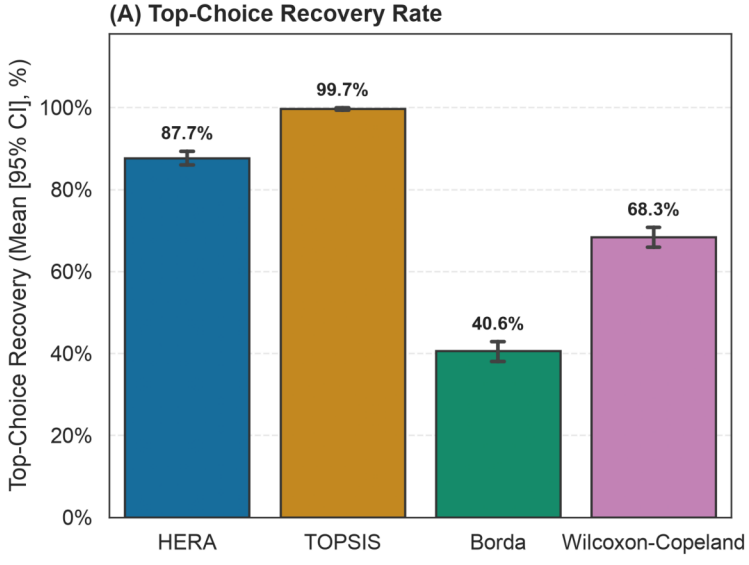
Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	75.0%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	74.0%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	73.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C11 (Rank 11)	71.0%	75.0%	80.0%	Monotonic Baseline Step
C12 (Rank 12)	70.0%	75.0%	80.0%	Monotonic Baseline Step
C13 (Rank 13)	69.0%	75.0%	80.0%	Monotonic Baseline Step
C14 (Rank 14)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

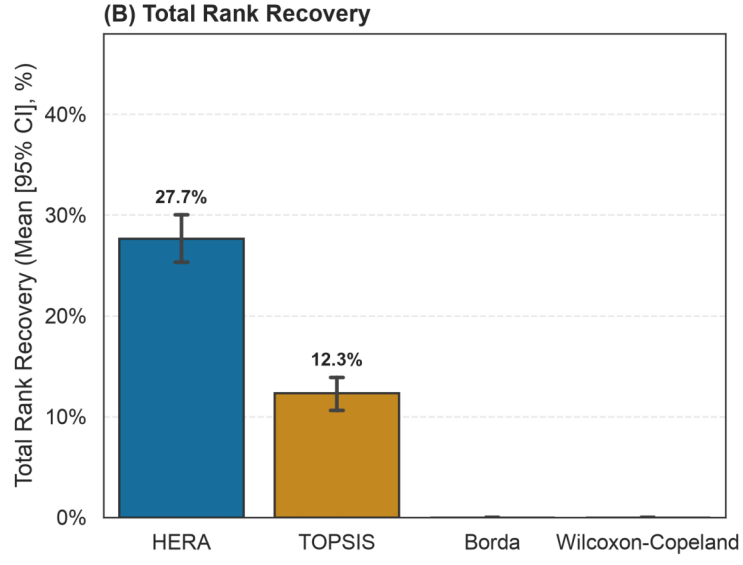

Ground-Truth Candidate Profiles
Candidate input presentation order was randomly permuted across trials to eliminate positional bias
Candidate Profiles (N = 10)
Candidate Profiles (N = 12)
Candidate Profiles (N = 14)
Metric Level (Median [IQR], %)
100
90
80
70
60
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
Candidates in Ground-Truth Order (Rank 1 to N)
M1 (Primary Metric)
M2 (Secondary Metric)
M3 (Tertiary Metric)

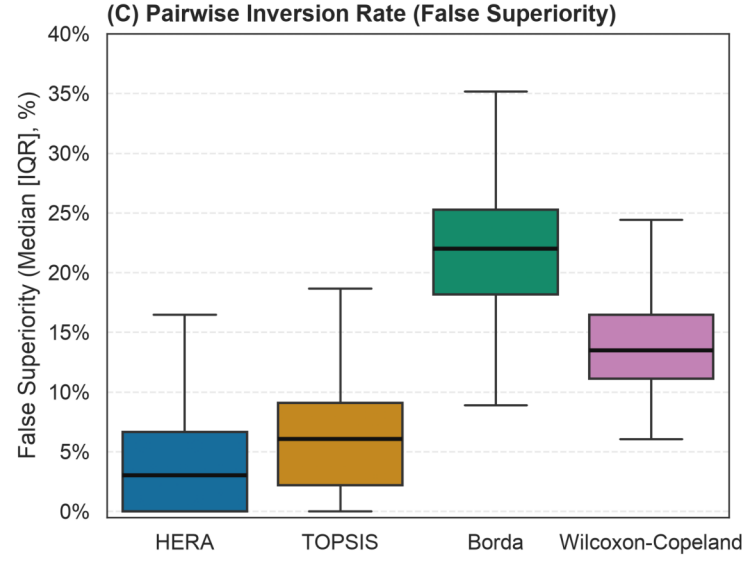

Benchmark Validation Overview
Default parameters (unless otherwise specified): N = 12, n = 50, Cliff's d ≈ 0.50, ρ = 0.0, K = 100 runs per point
(A) Top-Choice Recovery vs. Noise
Top-Choice Recovery (Mean [95% CI], %)
Noise Level (σ, %)
(B) Total Rank Recovery vs. Noise
Total Rank Recovery (Mean [95% CI], %)
Noise Level (σ, %)
(C) Pairwise Inversion Rate (FSR) vs. Noise
False Superiority (Median [IQR], %)
Noise Level (σ, %)
(D) Monotonic Alignment (Spearman's ρ) vs. Noise
Spearman's ρ (Median [IQR])
Noise Level (σ, %)
(E) Rank Fidelity vs. Sample Size
Kendall's τ (Median [IQR])
Cohort Sample Size (n)
(F) Scalability across Candidate Scale (N)
Kendall's τ (Median [IQR])
Number of Candidates (N)
(G) Rank Fidelity vs. Effect Size Magnitude
Kendall's τ (Median)
Small (d ≈ 0.25)
Medium (d ≈ 0.50)
Large (d ≈ 0.80)
Effect Size (Cliff's d)
(H) Robustness under Metric Collinearity
Kendall's τ (Median)
Independent (ρ = 0.0)
Correlated (ρ = 0.5)
Inter-Metric Correlation (ρ)
HERA
TOPSIS
Borda
Wilcoxon-Copeland

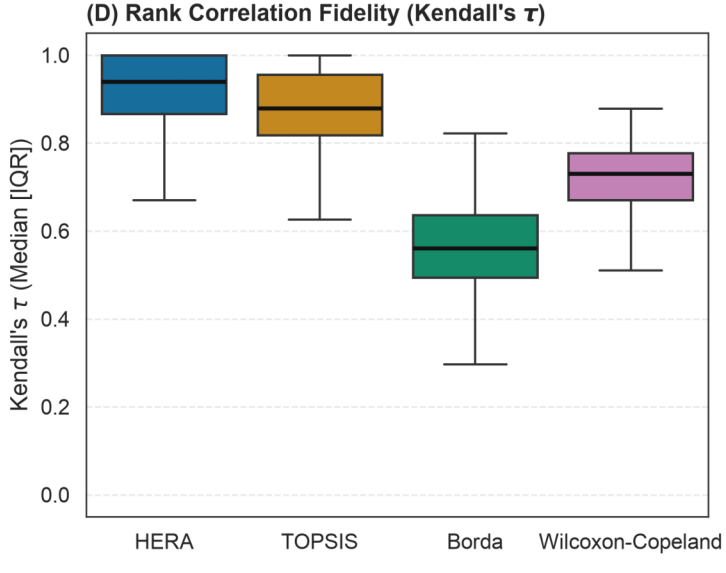
Pooled Distribution across Core Conditions

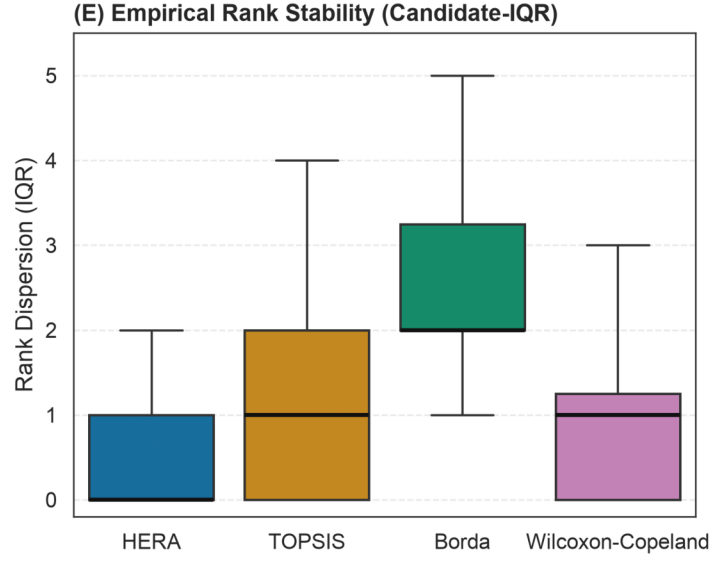
N ∈ {10, 12, 14}, σ ∈ {2%, 4%, 6%, 8%, 10%}, n = 50, ρ = 0.0; K = 1500 runs per method

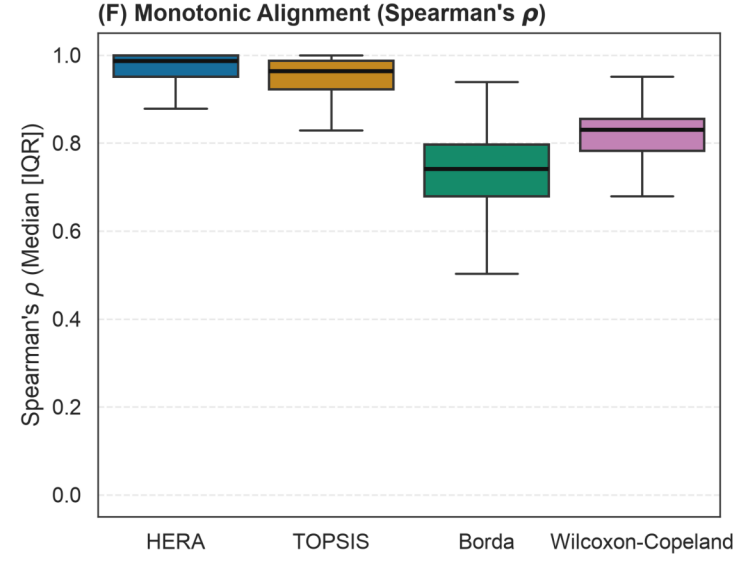


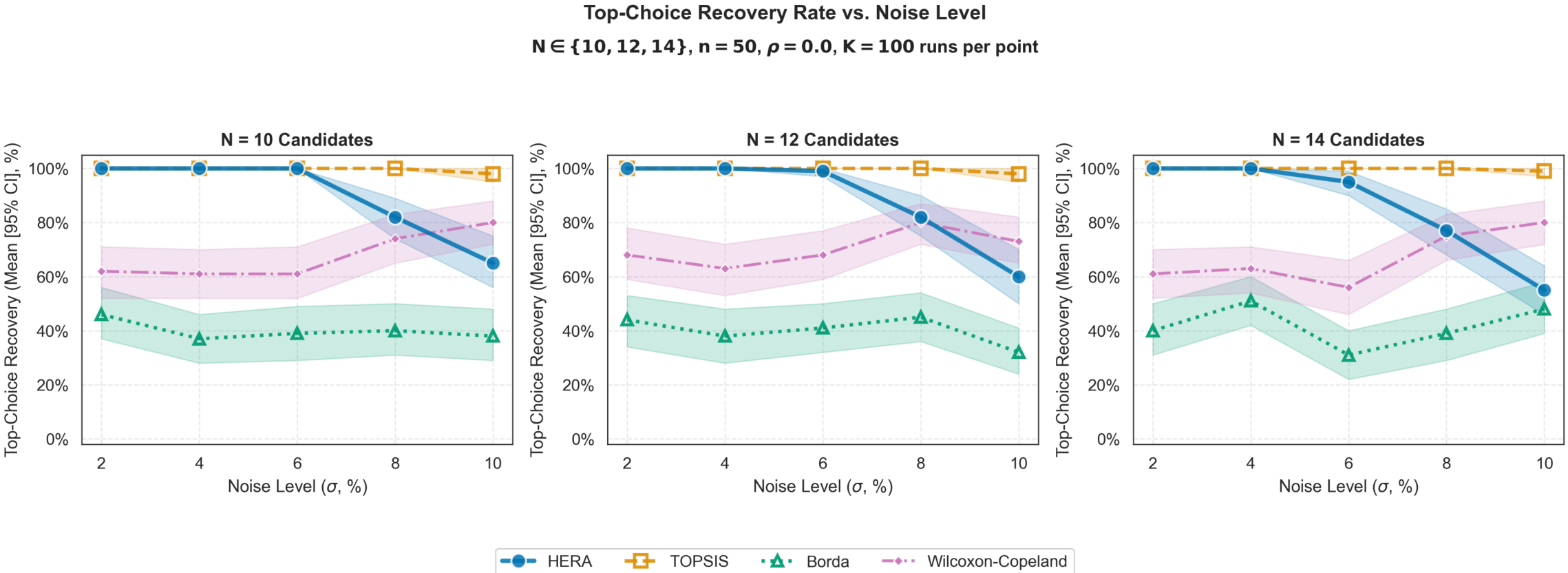


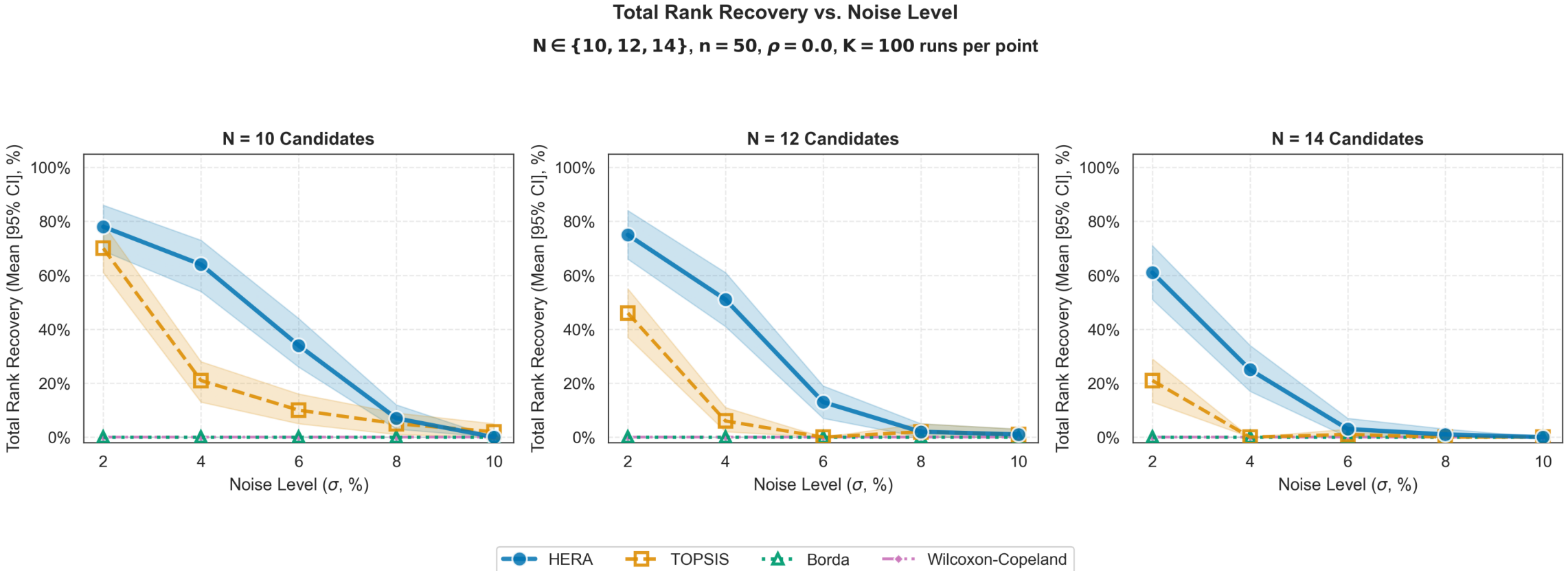


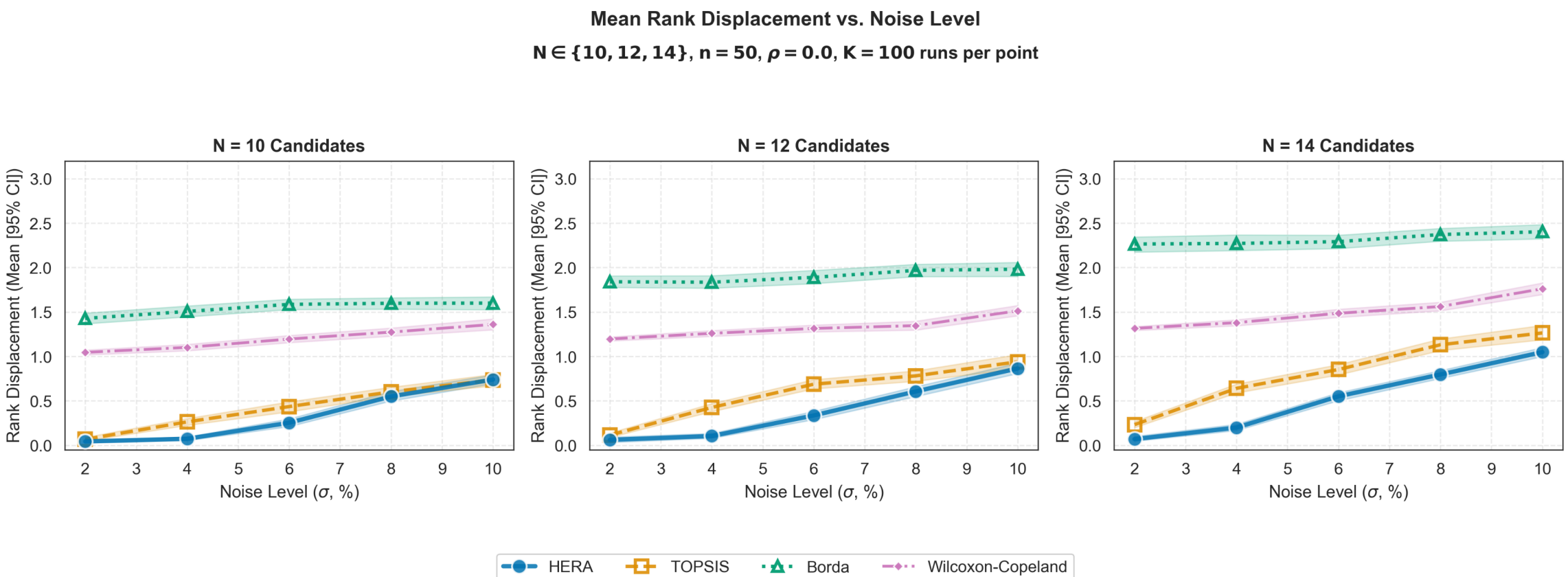


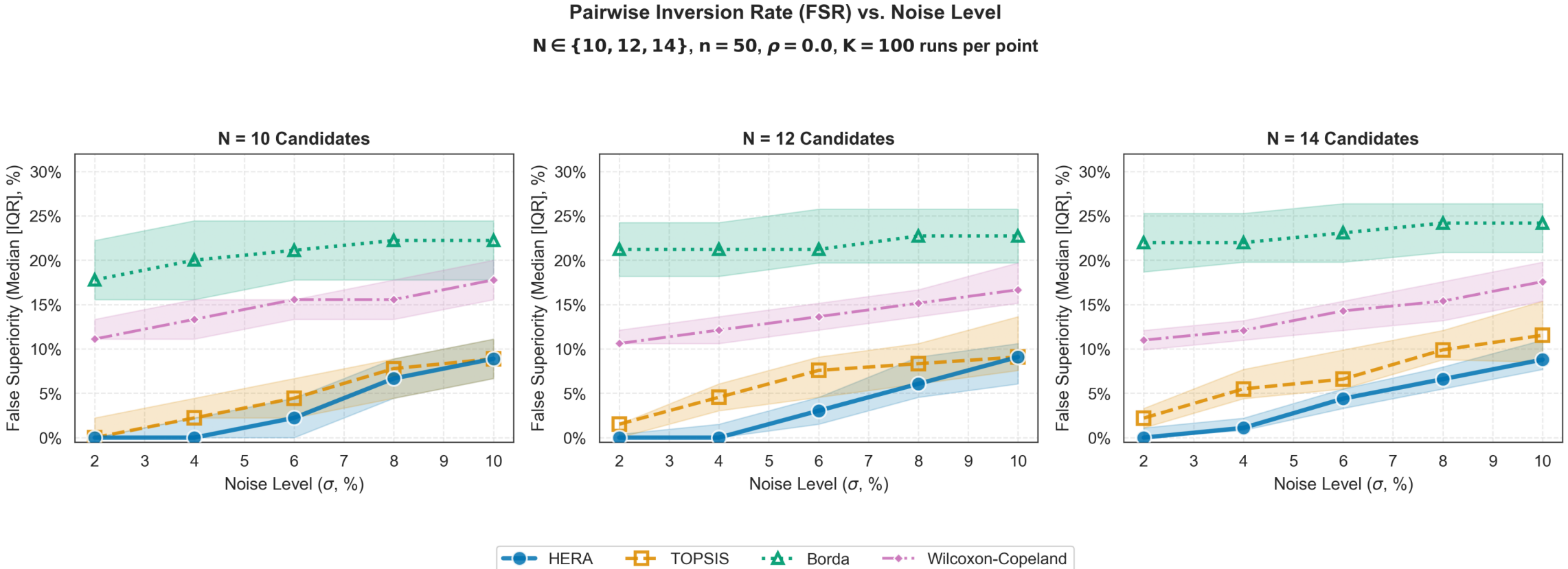


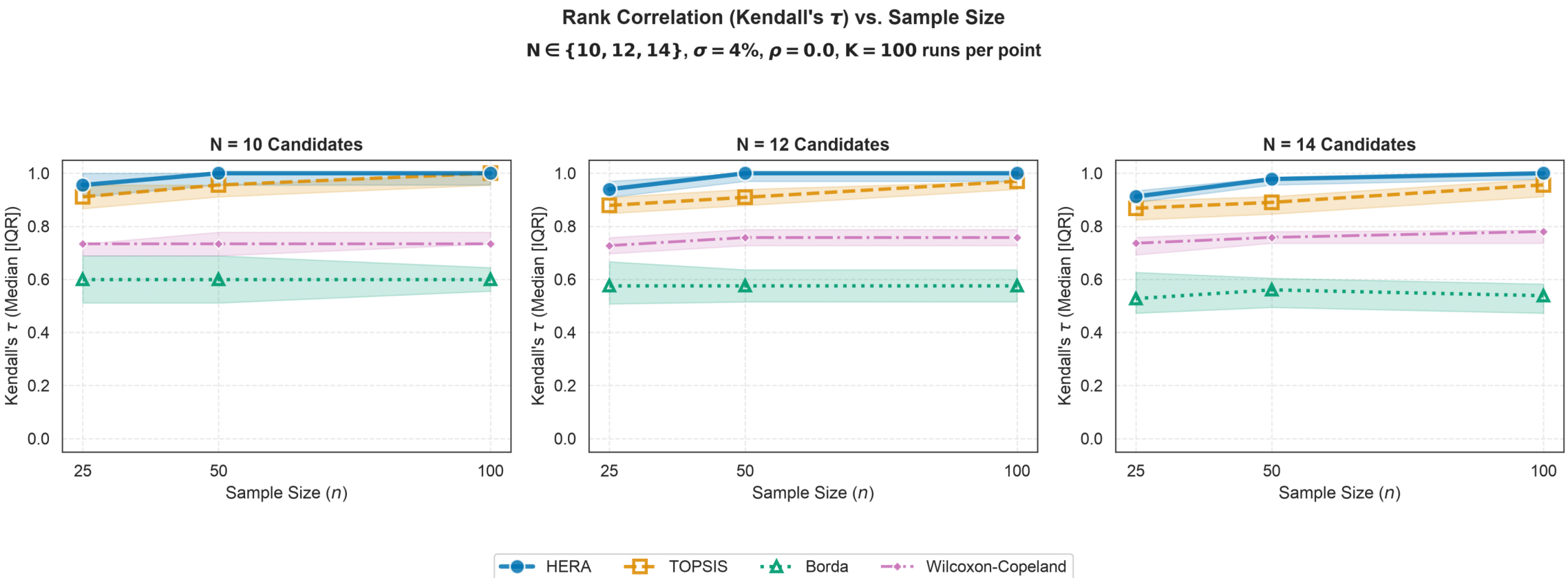


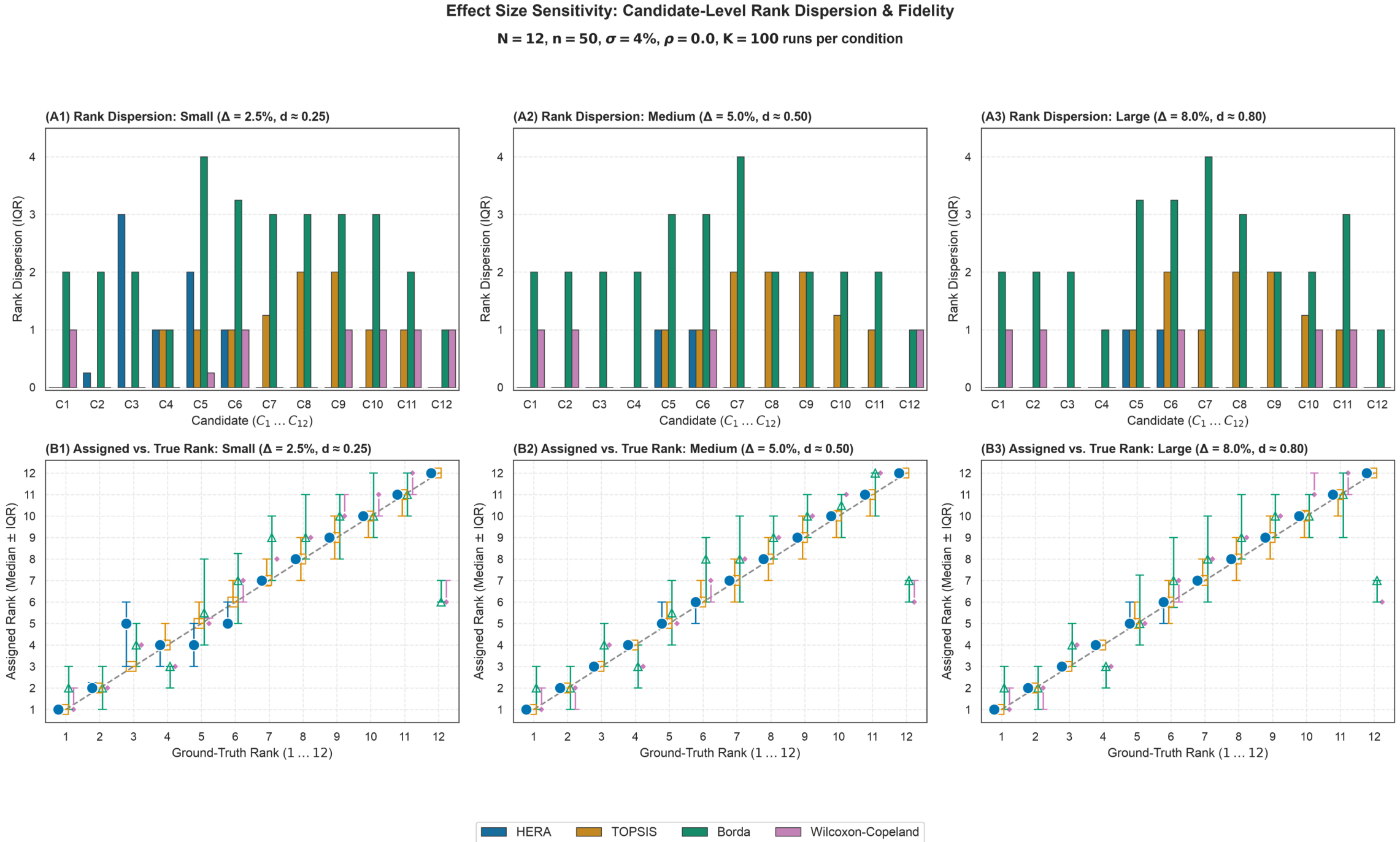

Top-Choice Recovery Rate vs. Noise Level
N ∈ {10, 12, 14}, n = 50, ρ = 0.0, K = 100 runs per point
N = 10 Candidates
N = 12 Candidates
N = 14 Candidates
Top-Choice Recovery (Mean [95% CI], %)
100%
80%
60%
40%
20%
0%
2
4
6
8
10
Noise Level (σ, %)
HERA
TOPSIS
Borda
Wilcoxon-Copeland


Total Rank Recovery vs. Noise Level
N ∈ {10, 12, 14}, n = 50, ρ = 0.0, K = 100 runs per point
N = 10 Candidates
N = 12 Candidates
N = 14 Candidates
Total Rank Recovery (Mean [95% CI], %)
100%
80%
60%
40%
20%
0%
2
4
6
8
10
Noise Level (σ, %)
HERA
TOPSIS
Borda
Wilcoxon-Copeland


Mean Rank Displacement vs. Noise Level
N ∈ {10, 12, 14}, n = 50, ρ = 0.0, K = 100 runs per point
N = 10 Candidates
N = 12 Candidates
N = 14 Candidates
Rank Displacement (Mean [95% CI])
Noise Level (σ, %)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2
3
4
5
6
7
8
9
10
HERA
TOPSIS
Borda
Wilcoxon-Copeland

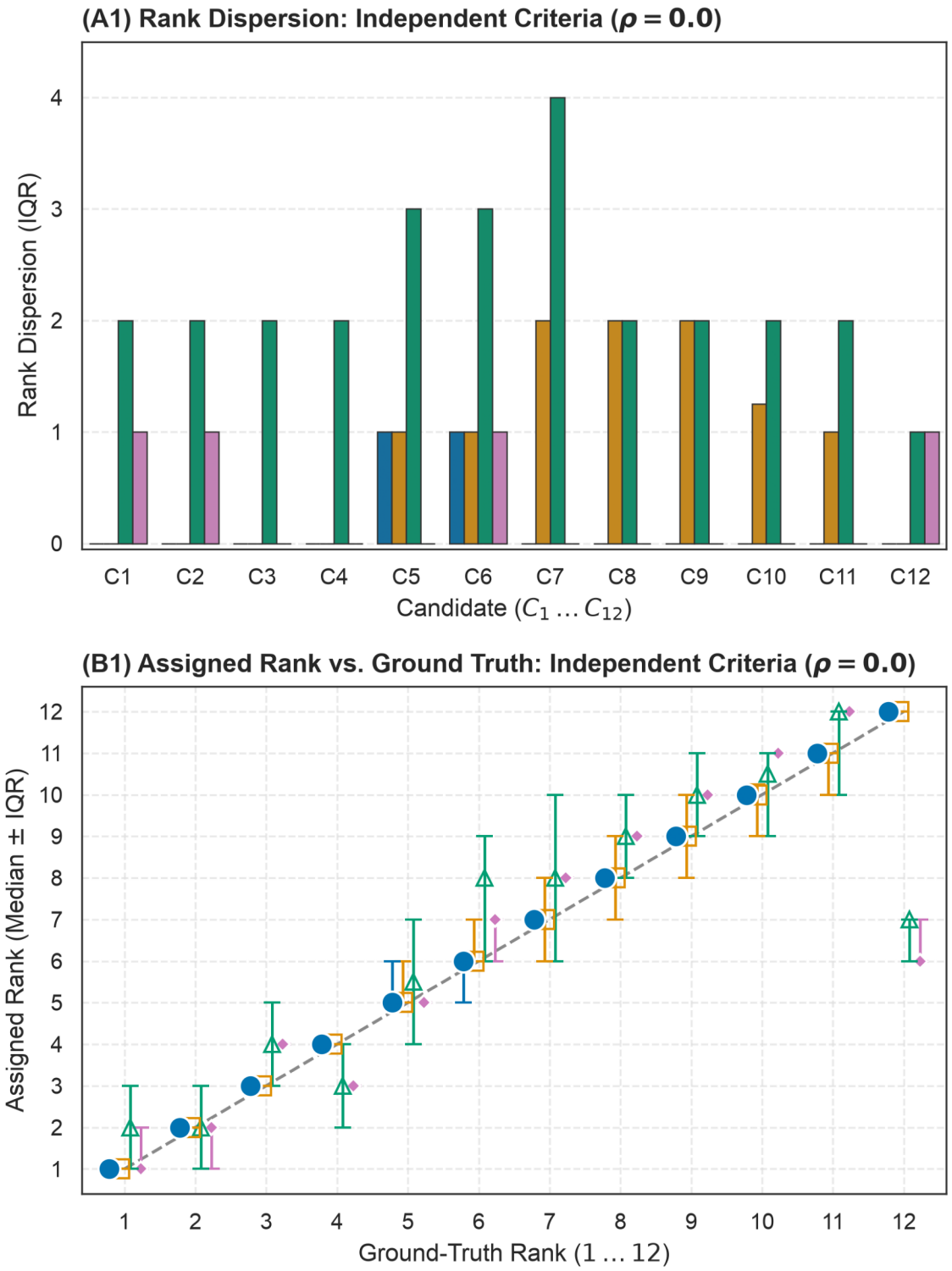

Pairwise Inversion Rate (FSR) vs. Noise Level
N ∈ {10, 12, 14}, n = 50, ρ = 0.0, K = 100 runs per point
N = 10 Candidates
N = 12 Candidates
N = 14 Candidates
False Superiority (Median [IQR], %)
30%
25%
20%
15%
10%
5%
0%
2
3
4
5
6
7
8
9
10
Noise Level (σ, %)
HERA
TOPSIS
Borda
Wilcoxon-Copeland

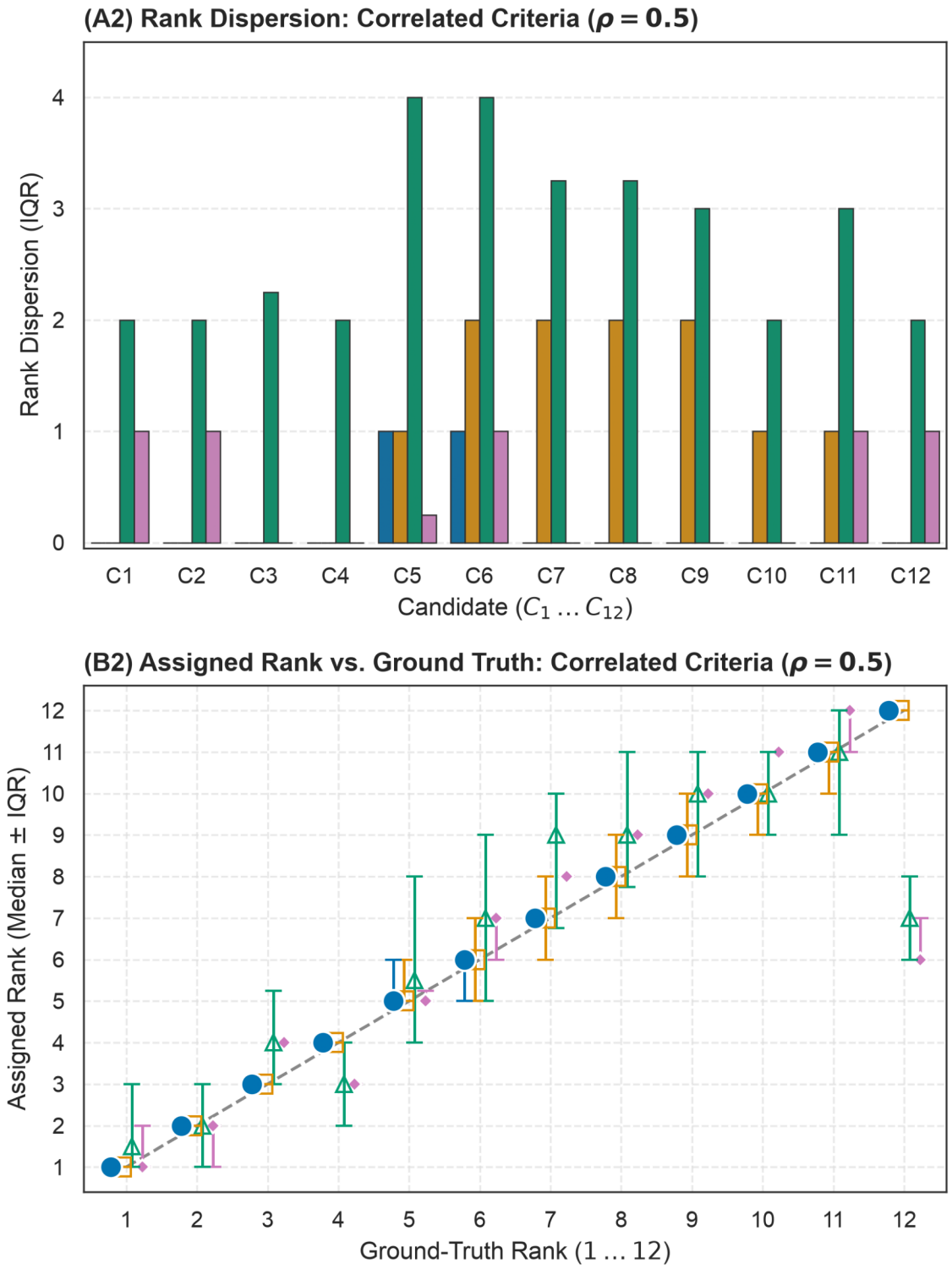

Rank Correlation (Kendall's τ) vs. Sample Size
N ∈ {10, 12, 14}, σ = 4%, ρ = 0.0, K = 100 runs per point
N = 10 Candidates
N = 12 Candidates
N = 14 Candidates
Kendall's τ (Median [IQR])
1.0
0.8
0.6
0.4
0.2
0.0
25
50
100
Sample Size (n)
HERA
TOPSIS
Borda
Wilcoxon-Copeland


Effect Size Sensitivity: Candidate-Level Rank Dispersion & Fidelity
N = 12, n = 50, σ = 4%, ρ = 0.0, K = 100 runs per condition
(A1) Rank Dispersion: Small (Δ = 2.5%, d ≈ 0.25)
(A2) Rank Dispersion: Medium (Δ = 5.0%, d ≈ 0.50)
(A3) Rank Dispersion: Large (Δ = 8.0%, d ≈ 0.80)
Rank Dispersion (IQR)
Candidate ($C_1 \dots C_{12}$)
C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12
(B1) Assigned vs. True Rank: Small (Δ = 2.5%, d ≈ 0.25)
(B2) Assigned vs. True Rank: Medium (Δ = 5.0%, d ≈ 0.50)
(B3) Assigned vs. True Rank: Large (Δ = 8.0%, d ≈ 0.80)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 12)
HERA
TOPSIS
Borda
Wilcoxon-Copeland

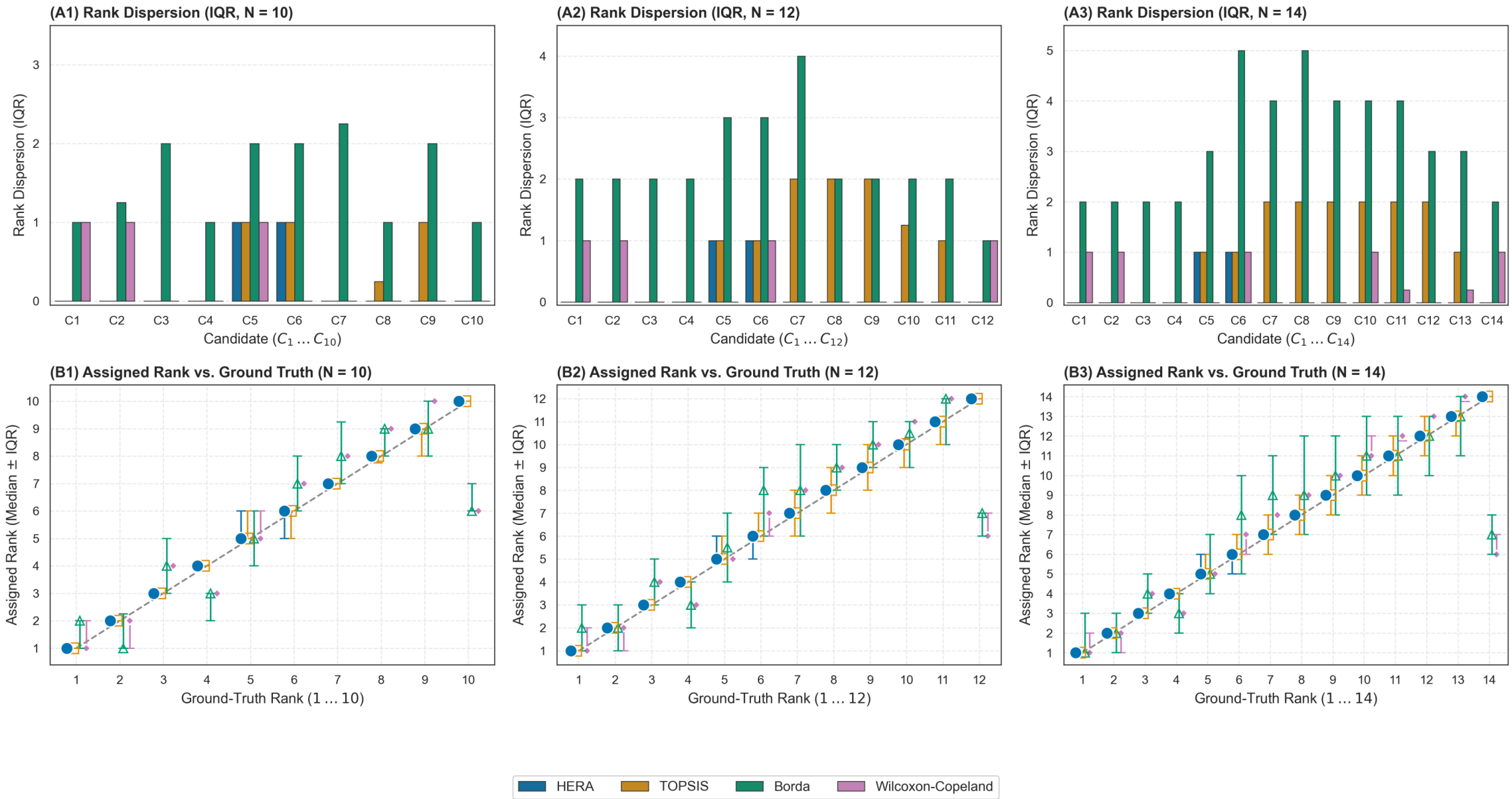
Inter-Metric Collinearity Sensitivity: Rank Dispersion & Fidelity

N = 12, n = 50, σ = 4%, Cliff's d ≈ 0.50, K = 100 runs per condition





HERA TOPSIS Borda Wilcoxon-Copeland


Candidate-Level Rank Stability Across Scales
N ∈ {10, 12, 14}, n = 50, σ = 4%, ρ = 0.0, K = 100 runs per point
(A1) Rank Dispersion (IQR, N = 10)
Rank Dispersion (IQR)
Candidate (C_1 … C_{10})
(A2) Rank Dispersion (IQR, N = 12)
Rank Dispersion (IQR)
Candidate (C_1 … C_{12})
(A3) Rank Dispersion (IQR, N = 14)
Rank Dispersion (IQR)
Candidate (C_1 … C_{14})
(B1) Assigned Rank vs. Ground Truth (N = 10)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 10)
(B2) Assigned Rank vs. Ground Truth (N = 12)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 12)
(B3) Assigned Rank vs. Ground Truth (N = 14)
Assigned Rank (Median ± IQR)
Ground-Truth Rank (1 … 14)
HERA
TOPSIS
Borda
Wilcoxon-Copeland